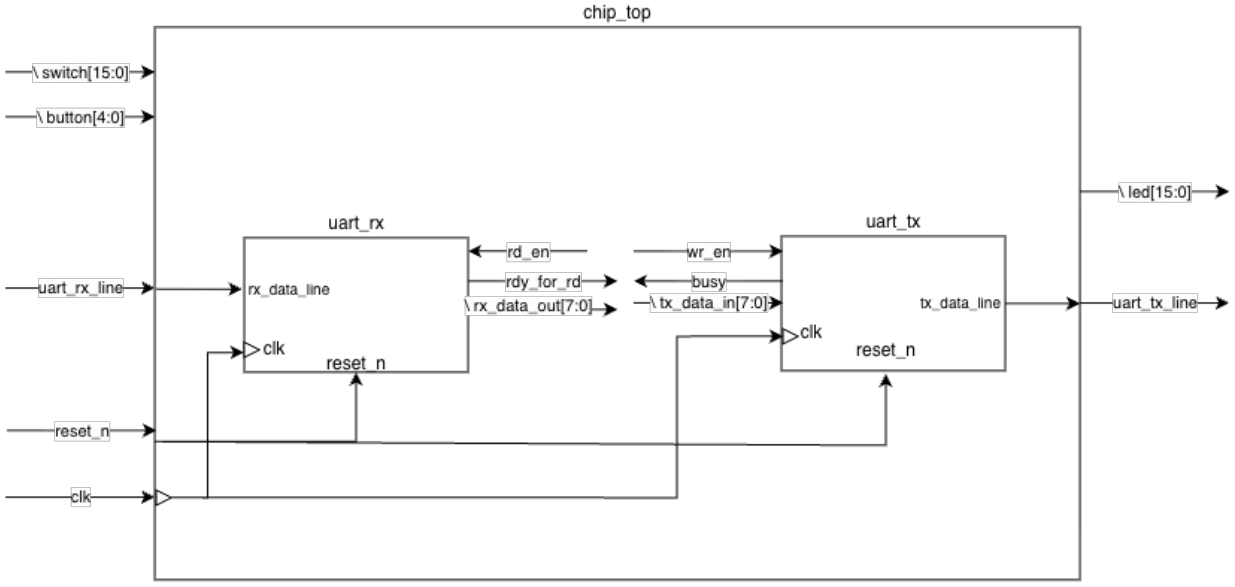
Lab 1 – Yuval Zecharia

.1



החיבורים של uart_tx ו-uart_rx צריכים להיות מחוברים לאיזה FSM ששולט מתי לשדר, לכתוב ואיזה מידע לשים על הקווים. **הם אינם מחוברים אחד לשני.**

טבלה chip_top:

Port name	Direction	Type	Description
clk	Input	Wire	Clk input to the chip
reset_n	Input	Wire	a/sync reset to the chip (Active low)
switch	Input	Wire [15:0]	Inputs from board switches
button	Input	Wire [4:0]	Inputs from board buttons
uart_rx_line	Input	Wire	Rx serial line to the chip
uart_tx_line	Output	Wire	Tx serial line from the chip
led	Output	Wire [15:0]	Led outputs from chip to the board

טבלה uart_rx:

Port name	Direction	Type	Description
clk	Input	Wire	Clock input from chip
reset_n	Input	Wire	General reset from chip (Active low)
rx_data_line	Input	Wire	Serial data in from chip_top
rd_en	Input	Wire	Part of handshake protocol. High when external consumer is ready for new word.
rdy_for_rd	Output	Wire	Part of handshake protocol. High ‘1’ when ready for a new word.
rx_data_out	Output	Wire [7:0]	Parallel data out from uart_rx module

טבלה uart_tx:

Port name	Direction	Type	Description
clk	Input	Wire	Clock input from chip
reset_n	Input	Wire	General reset from chip (Active low)
tx_data_in	Input	Wire [7:0]	Parallel in word data to transmit
wr_en	Input	Wire	Part of handshake protocol. High when data_in is valid and ready to transmit
busy	Output	Wire	Part of handshake protocol. Indicates new data is not ready to be loaded.
tx_data_line	Output	Wire	Serial transmission line.

Code:

chip_top.sv

```
`timescale 1ns / 1ps

module chip_top(
    input clk, reset_n,
    input uart_rx_line,
    input [15:0] switch,
    input [4:0] button,
    output uart_tx_line,
    output [15:0] led
    );

    wire [7:0] rx_data, tx_data;
    wire rd_en, rdy_for_rd;
    wire wr_en, busy;

    uart_rx uart_rx_i (
        .clk(clk),
        .reset_n(reset_n),
        .rx_data_line(uart_rx_line),
        .rd_en(rd_en),
        .rdy_for_rd(rdy_for_rd),
        .rx_data_out(rx_data)
    );

    uart_tx uart_tx_i(
        .clk(clk),
        .reset_n(reset_n),
        .tx_data_in(tx_data),
        .wr_en(wr_en),
        .busy(busy),
        .tx_data_line(uart_tx_line)
    );

endmodule
```

uart_rx.sv

```
`timescale 1ns / 1ps

module uart_rx(
    input clk, reset_n,
    input rx_data_line, rd_en,
    output rdy_for_rd,
    output [7:0] rx_data_out
    );

endmodule
```

uart_tx.sv

```
`timescale 1ns / 1ps

module uart_tx(
    input clk, reset_n,
    input [7:0] tx_data_in,
    input wr_en,
    output busy,
    output tx_data_line
    );
endmodule
```

Lint schematics:

